

SEMESTER 1

LOG #1 20/03/2024

- Had scheduling issues and was put into team Week 3.
 - Meeting with the team, client and mentor for the first time.

LOG #2 20/03/2024

- After meeting with mentor and client, team had a meeting.
 - Allocated tasks (from meeting minutes -> team meeting):

#	Action Item Description	Owner	Expected Completion Date
Project Proposal			
1.	Rationale	Thomas, Win	31/03/2025
2.	Objectives/Scope	Thomas, Win	31/03/2025
3.	Skills Analysis	Charmi, Thomas	31/03/2025
4.	Team Schedule	Kylie, Win	31/03/2025
5.	Project Management/Methodology	Zaf, Thomas	31/03/2025
6.	Risk Register	Larissa, Zaf	31/03/2025
7.	Issue Register	Larissa, Zaf	31/03/2025
8.	Work Breakdown Structure	Win, Thomas, Zaf	31/03/2025
9.	Project Estimated Cost	Larissa, Charmi	31/03/2025
10.	Project Feasibility	Larissa, Kylie	31/03/2025
11.	Gantt Chart	Nathan, Win	31/03/2025
12.	Terms of Reference	Charmi, Kylie	31/03/2025

LOG #3 20/03/2025

Notes from the workshop

Methodology breaks down a project into:

- Phases, tasks and sub-tasks -> using a work breakdown project (WBS).

Planning phase:

- Define scope, meeting clients, and creating a proposal.

Proposal = baseline for project execution

Different methods that can be used = scrum, XP, waterfall, RUP, UCD, SECS DLC.

Choice of methodology can depend on the client's preferences.

Practices that are important -> regular meetings, testing throughout all phases, consistent quality focus

Revise team contract – align this with chosen methodology/teamwork practices.

Proposal = must show that project is necessary, feasible and manageable.

LOG #4 22/03/2025

- As I was a late addition to the group, I reviewed the proposal to understand what is needed in our proposal.
- Focused on understanding the project's objective and what skills are needed to complete this project.

Notes I've written down when reviewing the proposal:

- Need to familiarise myself with Linux.
 - Networking (ipv4, ipv6, TCP/UDP).
 - Performance testing tools (iPerf/DITG).
 - Analysing data performance metrics
- Project objective -> need to evaluate IPv4 & 6 and their network performance using three linux based software.
- Why: IPv4 has scalability issues -> IPv6 is considered a replacement

LOG #5 23/03/2025

Updated my logbook.

LOG #6 24/03/2025:

Missed week 1 – had to do some catchup.

Notes from lecture slides:

- RND runs through both semesters.
- Purpose: apply academic skills to solve a real client problem through a functional prototype.
- Expectations: teamwork, off-site hours, workshops and supervisor meetings.
- Worklog book and team portfolio are essential -> records the process of the project and individual work.
- Taking initiative, showing respect and maintaining professionalism is important.
- Project proposal is due Week 6 (10%).

LOG #7 25/03/2025:

Missed week 2 – had to do some catchup.

Notes from lecture slides:

- Project proposal = proves it is necessary, feasible and manageable.

- Proposal should have = background info, project rationale, objectives, scope and a clear work breakdown structure (WBS).
- Client fact finding = very important -> helps identify business goals, IT issues and opportunities.
- Planning upskilling schedules -> find skills you lack on -> record everything in team portfolio.
- Proposal should be concise and should be under 10 pages -> detailed plans in appendix.
- Prepare client meetings professionally -> allocating roles and responsibilities (e.g. minute-taker, question-asker etc).

LOG #8 25/03/2025

Upskilling: watched an introduction on linux

<https://www.youtube.com/watch?v=BMGixkvJ-6w>

Key commands needed:

- Pwd = (printing working directory) shows the current directory
- Ls = lists all files and folders in the current directory
- Cd = change directory
- Mkdir = new folder
- Rmdir = removes empty folder
- Touch = new file
- Rm = deletes file
- Cp = copies folders/folders
- Mv = moves or renames folders
- Cat = displays contents in a file
- Clear = clears the terminal
- Chmod = change permissions
- Sudo = admin access
- ../ = goes back one level
- Apt install = package manger that installs software.

LOG #9 27/03/2025

Attended mentor meeting (in meeting minutes)

- Project update and reviewing the proposal

LOG #10 27/03/2025

Attended workshop.

Notes from the lecture:

Team culture = how members work together, treat each other and share values.

Team values = respect, diversity, communication, trust, accountability and learning continuously.

Tuckman's stages of team development = forming, storming, norming, performing, and adjourning.

Conflict = can be both positive and negative. Positive -> can encourage new ideas. Negative -> causes stress and mistrust.

Conflict prevention:

- Create a safe space.
- Early intervention
- Communicating

Common conflicts:

- Poor attendance
- Missing deadlines
- Dominating members
- Lack of effort

NO BLAMING OR SCAPEGOATING!! -> instead have constructive discussions and as a group create solutions.

Proposal presentation = DUE NEXT WEEK

Quality assurance = DO THIS BEFORE SUBMISSION NEXT WEEK

LOG #11 28/03/2025

Drafted the project feasibility section – also reviewed the proposal:

Project Feasibility: The proposal includes an analysis of the project's feasibility. Information included should consider the required infrastructure, tools, technologies, roles, and skills. The report must include evidence of alternatives that were considered and rationale/justification for the path ultimately taken.

Note: Even if the project is a continuation of an existing project, the feasibility of the next phase and the proposed activities within the next phase must be evaluated.

Required Infrastructure

Hardware:

Software: Fedora OS, Ubuntu and a chose third Linux-based OS.

LOG #12 30/03/2025

Updated my logbook.

LOG #13 30/03/2025

Team meeting online (in meeting minutes)

- Discussed about our proposal and presentation.
- Reviewed feedback from mentor.

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	Modified by: Kylie Afable, Win Phyo, Larissa Goh and Charmi Patel	9:55 PM
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LOG #14 31/03/2025

Presentation = helped draft and picked the possible layout of the PowerPoint

Found in planning and control -> Proposal -> Proposal presentation -> version history -> first draft PowerPoint.

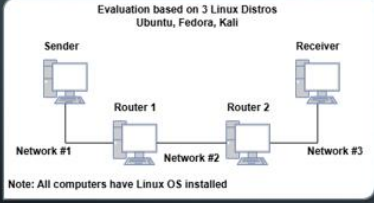
02 April 2025

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LOG #15 2/04/2025

PROJECT FEASIBILITY

Required Infrastructure:



Note: All computers have Linux OS installed

Tools and Technology:

- OS = Ubuntu, Fedora, Kali Linux
- Network Tools = iPerf (tests IPv4 and IPv6 using TCP and UDP)
- Packet Sizes = 12 variations of packet sizes

Required Skills Needed:

- Familiarity with Linux, software routing, and performance testing.
- IPv4 and IPv6 network principles
- Programming and debugging
- Router configuration

Worked on presentation – mainly project feasibility section

LOG #16 3/05/2025

Mentor/client meeting (in meeting minutes)

- Mock presentation and project update.

LOG #17 3/04/2025

Attended workshop

Notes:

Quality assurance & quality control

QA = prevents bugs -> following proper processes.

QC = detects bugs -> testing and reviews

Quality attributes = correctness, usability, reliability, and efficiency.

QA should be applied at every phase -> planning, design, and implementation

Must prepare a QA plan with walkthroughs, checklists and metrics.

03 April 2025

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	Modified by: Larissa Goh Open version	1:32 PM
	Modified by: Win Phyo and Larissa Goh Open version	12:57 PM
	Modified by: Win Phyo Open version	12:51 PM
	Modified by: Thomas Hugh Robinson and Kylie Afable Open version	12:02 PM
	Modified by: Thomas Hugh Robinson Open version	5:06 AM

PROPOSAL REQUIREMENTS:

- 10 min presentation, max 7 slides
- Everyone must speak/know what is in the proposal.
- Expectation = 25-30mins of questions and answers.
- Dress professionally.

Presentation should be proofread and uploaded by WEEK 5

After presentation -> update proposal based on feedback

Real project work will start after the presentation.

LOG #18 2/04/2025

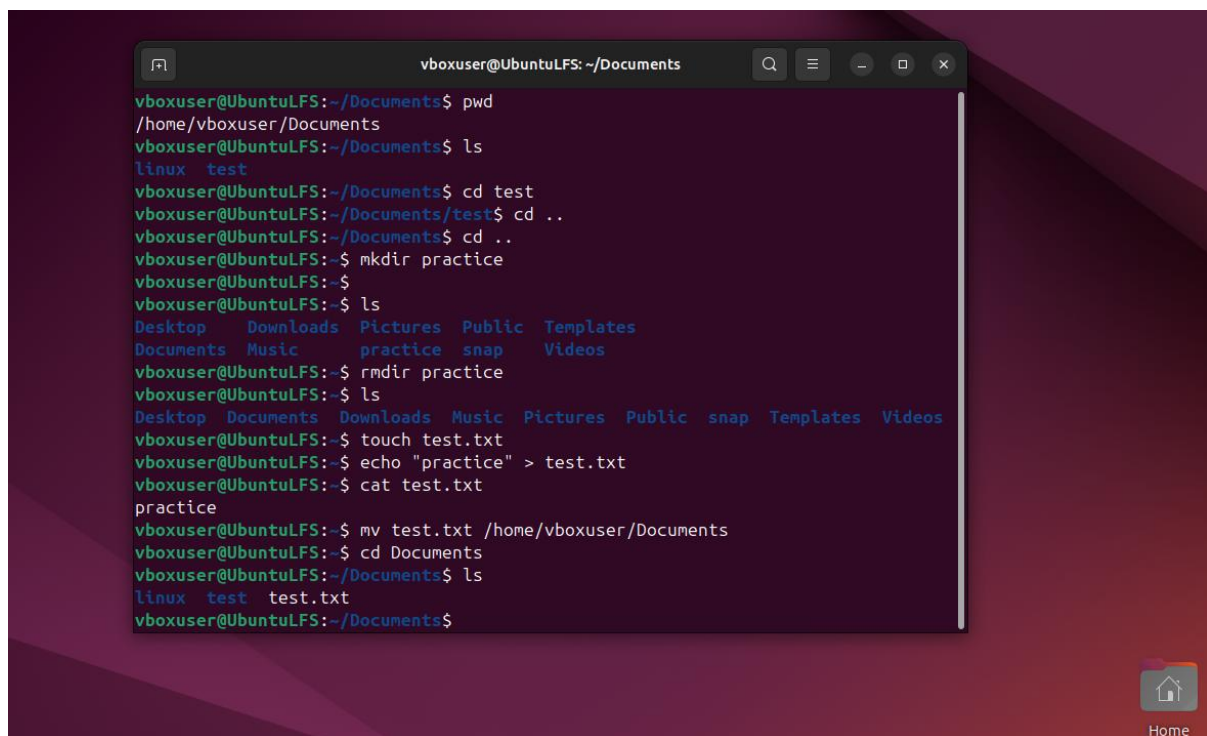
Upskilling – familiarising myself with Linux.

<https://www.youtube.com/watch?v=sWbUDq4S6Y8&t=1444s>

- 6-hour long tutorial on Linux.
- Had to break this video down in smaller bits
- Watched until chapter 4.

Image is from virtual box using Ubuntu.

Practiced basic Linux commands.

A screenshot of a terminal window titled 'vboxuser@UbuntuLFS: ~/Documents'. The terminal shows a series of Linux commands and their outputs. The user starts by running 'pwd', which returns '/home/vboxuser/Documents'. Then they run 'ls', showing 'linux test'. They navigate into the 'test' directory with 'cd test' and then back to the parent directory with 'cd ..'. Next, they create a new directory 'practice' using 'mkdir practice'. They run 'ls' again, showing the updated directory listing which includes 'practice'. Then they run 'rm -r practice' to remove it. Another 'ls' shows the directory is gone. They then create 'test.txt' with 'touch test.txt', write 'practice' into it with 'echo "practice" > test.txt', and verify the content with 'cat test.txt'. Finally, they move the file back to the root of the Documents directory with 'mv test.txt /home/vboxuser/Documents' and run 'ls' to confirm 'test.txt' is now in the root of the directory.

```
vboxuser@UbuntuLFS: ~/Documents
vboxuser@UbuntuLFS:~/Documents$ pwd
/home/vboxuser/Documents
vboxuser@UbuntuLFS:~/Documents$ ls
linux test
vboxuser@UbuntuLFS:~/Documents$ cd test
vboxuser@UbuntuLFS:~/Documents/test$ cd ..
vboxuser@UbuntuLFS:~/Documents$ cd ..
vboxuser@UbuntuLFS:~$ mkdir practice
vboxuser@UbuntuLFS:~$ ls
Desktop  Downloads  Pictures  Public  Templates
Documents Music      practice snap      Videos
vboxuser@UbuntuLFS:~$ rm -r practice
vboxuser@UbuntuLFS:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
vboxuser@UbuntuLFS:~$ touch test.txt
vboxuser@UbuntuLFS:~$ echo "practice" > test.txt
vboxuser@UbuntuLFS:~$ cat test.txt
practice
vboxuser@UbuntuLFS:~$ mv test.txt /home/vboxuser/Documents
vboxuser@UbuntuLFS:~$ cd Documents
vboxuser@UbuntuLFS:~/Documents$ ls
linux test test.txt
vboxuser@UbuntuLFS:~/Documents$
```

LOG #19 6/04/2025

Reviewed week 5 and updated logbook.

LOG #20 7/04/2025

Presentation rehearsal online (can be seen in team meeting minutes).

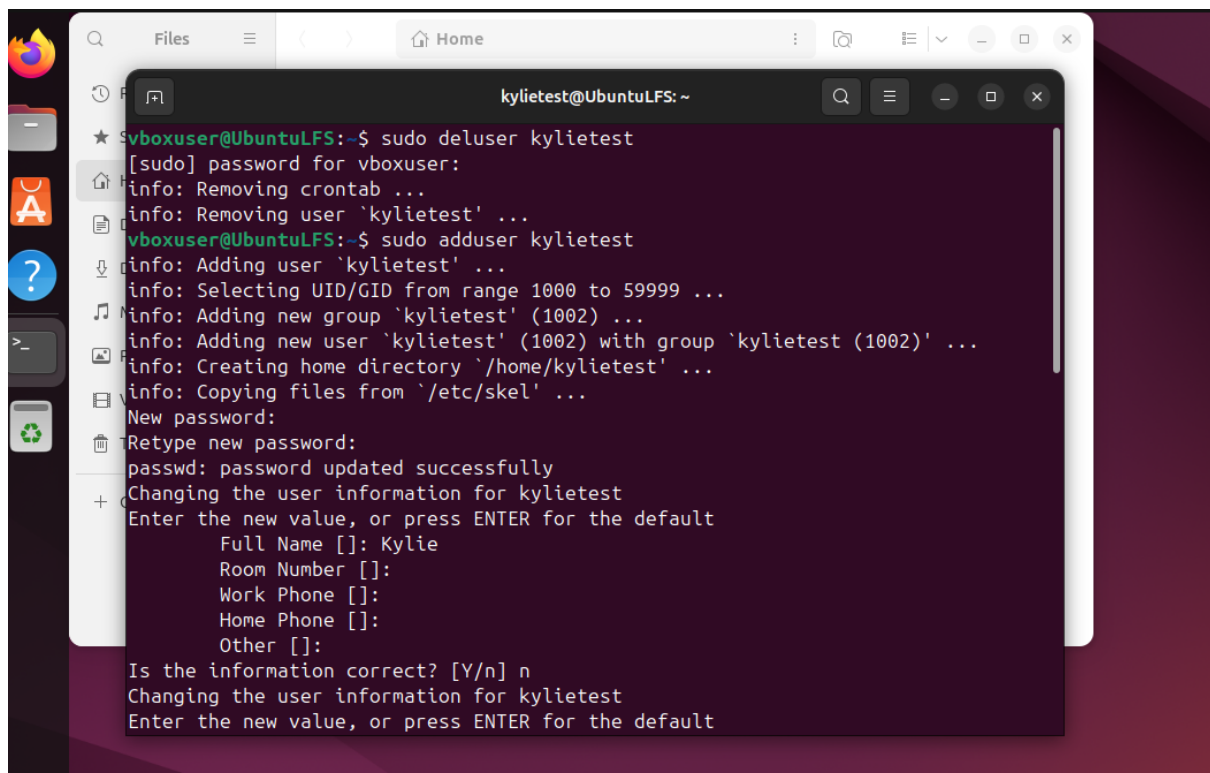
LOG #21 8/04/2025

Upskilling:

Continued off from this video: <https://www.youtube.com/watch?v=sWbUDq4S6Y8&t=1444s>

ALSO:

Practicing to add, switch and delete users



```
kylietest@UbuntuLFS: ~  
vboxuser@UbuntuLFS:~$ sudo deluser kylie  
[sudo] password for vboxuser:  
info: Removing crontab ...  
info: Removing user 'kylie' ...  
vboxuser@UbuntuLFS:~$ sudo adduser kylie  
info: Adding user 'kylie' ...  
info: Selecting UID/GID from range 1000 to 59999 ...  
info: Adding new group 'kylie' (1002) ...  
info: Adding new user 'kylie' (1002) with group 'kylie (1002)' ...  
info: Creating home directory '/home/kylie' ...  
info: Copying files from '/etc/skel' ...  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for kylie  
Enter the new value, or press ENTER for the default  
Full Name []: Kylie  
Room Number []:  
Work Phone []:  
Home Phone []:  
Other []:  
Is the information correct? [Y/n] n  
Changing the user information for kylie  
Enter the new value, or press ENTER for the default
```

Testing Networking commands:

```

kylietest@UbuntuLFS:~$ ping google.vom
ping: google.vom: Name or service not known
kylietest@UbuntuLFS:~$ ping google.com
PING google.com (142.250.66.206) 56(84) bytes of data.
64 bytes from syd09s23-in-f14.1e100.net (142.250.66.206): icmp_seq=1 ttl=255 time=30.7 ms
64 bytes from syd09s23-in-f14.1e100.net (142.250.66.206): icmp_seq=2 ttl=255 time=32.1 ms
64 bytes from syd09s23-in-f14.1e100.net (142.250.66.206): icmp_seq=3 ttl=255 time=32.9 ms
64 bytes from syd09s23-in-f14.1e100.net (142.250.66.206): icmp_seq=4 ttl=255 time=30.1 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3123ms
rtt min/avg/max/mdev = 30.114/31.459/32.886/1.097 ms
kylietest@UbuntuLFS:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=29.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=30.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=31.4 ms

```

Learning to install software and package management:

```

kylietest@UbuntuLFS:~$ sudo apt update
[sudo] password for kylietest:
Hit:1 http://nz.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://nz.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://nz.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://nz.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,543 kB]
Get:5 http://nz.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [292 kB]
Get:6 http://nz.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:7 http://nz.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.4 kB]
Get:8 http://nz.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,175 kB]
Get:9 http://nz.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [490 kB]

```

```

kylietest@UbuntuLFS:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  linux-headers-6.14.0-33-generic linux-hwe-6.14-headers-6.14.0-33
  linux-hwe-6.14-tools-6.14.0-33 linux-image-6.14.0-33-generic
  linux-modules-6.14.0-33-generic linux-modules-extra-6.14.0-33-generic
  linux-tools-6.14.0-33-generic
The following packages will be upgraded:
  bind9-dnsutils bind9-host bind9-libs bluez bluez-cups bluez-obexd cloud-init
  distro-info-data dpkg dpkg-dev firmware-sof-signed gdm3 ghostscript
  gir1.2-gdm-1.0 gir1.2-javascriptcoregtk-4.1 gir1.2-javascriptcoregtk-6.0
  gir1.2-nm-1.0 gir1.2-webkit-6.0 gir1.2-webkit2-4.1 libbluetooth3
  libdpkg-perl libgdm1 libgs-common libgs10 libgs10-common libgtop-2.0-11
  libgtop2-common libjavascriptcoregtk-4.1-0 libjavascriptcoregtk-6.0-1
  libldb2 libnm0 libnss-systemd libpam-systemd libpoppler-cpp0t64
  libpoppler-glib8t64 libpoppler134 libsmbclient0 libssl3t64 libsystemd-shared

```

LOG #22 8/04/2025

08 April 2025		
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Tools Comparison		
Comparison	iPerf	D-ITG
Performance	- Measures the maximum achievable bandwidth on IP networks. - Reports throughput, bitrate, loss and other parameters for each test. - Consists of features such as nuttcp and netperf. - Not backwards compatible with original iPerf.	- D-ITG measures delay, jitter, loss, throughput. - Uses logs to calculate performance metrics. - D-ITG comprises of different components: - ITGSend - ITGRecv - ITGLog - ITGManager - ITGDec
Feature 1	- Rewritten from scratch - Goal is to have a smaller, simpler code base. - Has a library version for other programs	- Supports layer 7 features: - Predefined stochastic profiles (e.g. Telnet, VoIP). - Custom payload (random or a file) - Both PS and IDT support stochastic processes.
Feature 2	- Measures bandwidth and performance - Supports TCP and UDP - Reports packet loss and delay	- Packet-level QoS metrics: - Bitrate, packet rate, one way delay, round trip time, jitter and packet loss.

Chosen Evaluation Network Tool: D-ITG

Edited the tools-comparison section – made sure to summarise to fit into the slides.

Also practiced for the presentation + timed how long my section was.

LOG #23 10/04/2025

MOCK PRESENTATION

LOG #24 10/04/2025

Updated logbook and reviewed week 6.

LOG #25 15/04/2025

Team meeting (refer to team meeting minutes):

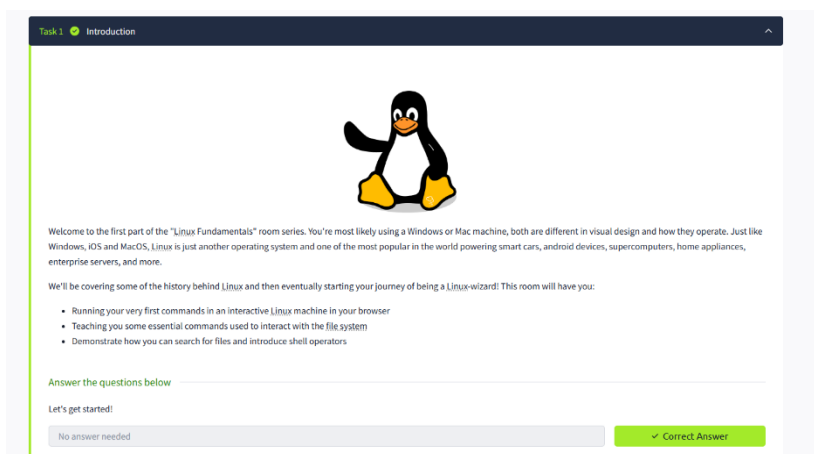
- Week 1 (current) focusing on Linux fundamentals and environment setup
- Week 2 Linux routing and network testing

LOG #26 16/04/2025

UPSKILLING: <https://www.youtube.com/watch?v=kPylihJRG70&t=1813s>

Part 2: <https://www.youtube.com/watch?v=7Zt2Mp2leBI>

Video had a free practice website for linux.



LOG #27 18/04/2025

Upskilling: continued from part 3 of previous videos.

<https://www.youtube.com/watch?v=bwgaZCb2ft8>

Notes:

Basic utility commands

- Wget = for downloading files
- Scp = secure copying between machines
- Python3 -m http.server = running a quick webserver.

Processes:

- Ps aux, top = list running processes.
- KILL, SIGTERM, SIGKILL, fg

Crontab -e = to schedule tasks (CRON)

Systemctl = to start/stop/enable services at boot

Apt update/install = adding or remove repositories

Viewing log files = /var/log

LOG #28 20/04/2025

Updated logbook.

LOG #29 21/04/2025

Linux routing and networking testing upskilling.

<https://www.youtube.com/watch?v=vDWY3PuHMX8> = has networking commands for beginners

commands I learnt from researching on routing and networking testing

- ifconfig and ifconfig -a = show all interfaces + inactive ones
- ip addr show, ip route show, ip -br addr = shows each interface and IPs.
- Netstat -r = shows routing table
- Netstat -tuln = shows TCP/UDP ports that are in use
- Netstat -a = shows all the connections
- Curl google.com = shows the html of google homepage
- Curl -I (<https://pretend.com>) = shows only response headers
- "" -o pretend.html pretend.com = downloads and saves to a file
- Ping (enter here) = keeps pinging until you want it to stop
- Ping -c 4 = sends only 4 packets
- Ping -6 = test ipv6 connection

LOG#30 22/04/2025

Team meeting – reviewing what we have done for week 1 upskilling.

LOG #31 24/04/2025

Upskilling routing and network testing.

Practiced some of the commands on ubuntu virtual box.

```
kyltetest@UbuntuLFS:~$ whoami
kyltetest
kyltetest@UbuntuLFS:~$ pwd
/home/kyltetest
kyltetest@UbuntuLFS:~$ ls -l
total 36
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Desktop
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Documents
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Downloads
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Music
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Pictures
-rwx----- 1 kyltetest kyltetest 0 Oct 28 05:01 practice.txt
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Public
drwx----- 3 kyltetest kyltetest 4096 Oct 28 06:25 snap
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Templates
drwxr-xr-x 2 kyltetest kyltetest 4096 Oct 28 06:25 Videos
kyltetest@UbuntuLFS:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:3eff:2a45:8088:cefa prefixlen 64 scopeid 0x0<gl
    obal>
    inet6 fd17:625c:f037:2:a00:27ff:fe19:ad61 prefixlen 64 scopeid 0x0<glo
    bal>
    inet6 fe80::a00:27ff:fe19:ad61 prefixlen 64 scopeid 0x20<link>
```

```
kyltetest@UbuntuLFS:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
    oup default qlen 1000
    link/ether 08:00:27:19:ad:61 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86308sec preferred_lft 86308sec
    inet6 fd17:625c:f037:2:3eff:2a45:8088:cefa scope global temporary dynamic
        valid_lft 86310sec preferred_lft 14310sec
    inet6 fd17:625c:f037:2:a00:27ff:fe19:ad61/64 scope global dynamic mngtppadr
        valid_lft 86310sec preferred_lft 14310sec
    inet6 fe80::a00:27ff:fe19:ad61/64 scope link
        valid_lft forever preferred_lft forever
kyltetest@UbuntuLFS:~$ curl google.com
Command 'curl' not found, but can be installed with:
sudo snap install curl # version 8.16.0, or
```

```
[sudo] password for kyltetest:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  liblvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 226 kB of archives.
After this operation, 534 kB of additional disk space will be used.
Get:1 http://nz.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.
5.0-2ubuntu10.6 [226 kB]
Fetched 226 kB in 0s (1,249 kB/s)
Selecting previously unselected package curl.
(Reading database ... 195155 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.6) ...
Setting up curl (8.5.0-2ubuntu10.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
kyltetest@UbuntuLFS:~$
```

```
liblvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 226 kB of archives.
After this operation, 534 kB of additional disk space will be used.
Get:1 http://nz.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.
5.0-2ubuntu10.6 [226 kB]
Fetched 226 kB in 0s (1,249 kB/s)
Selecting previously unselected package curl.
(Reading database ... 195155 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.6) ...
Setting up curl (8.5.0-2ubuntu10.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
kyltetest@UbuntuLFS:~$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
kyltetest@UbuntuLFS:~$
```

```

top - 06:28:19 up 3 min, 1 user, load average: 0.42, 0.50, 0.27
Tasks: 224 total, 2 running, 222 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.4 sy, 0.0 ni, 99.2 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 2926.0 total, 1021.9 free, 1049.8 used, 1041.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 1076.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S   %CPU  %MEM    TIME+  COMMAND
 1972 kyltetest 20   0 4130576 287460 127832 R    1.7   9.6   0:13.97 gnome-s+
 501  root      20   0      0      0      0    I    0.2   0.0   0:00.08 kworker+
 878  root      20   0   18148   8732   7708 S    0.2   0.3   0:00.35 systemd+
 2179 kyltetest 20   0 388844   12300   7052 S    0.2   0.4   0:01.37 ibus-da+
 2494 kyltetest 20   0 236768   7224   6584 S    0.2   0.2   0:00.34 ibus-en+
 2633 kyltetest 20   0 2923824 205652 118728 S    0.2   6.9   0:01.49 gnome-l+
 2813 kyltetest 20   0 553636 52580 41804 S    0.2   1.8   0:01.98 gnome-t+
    1  root      20   0   23256  14164   9428 S    0.0   0.5   0:23.32 systemd
    2  root      20   0      0      0      0    S    0.0   0.0   0:00.01 kthreadd
    3  root      20   0      0      0      0    S    0.0   0.0   0:00.00 pool_wor
    4  root      0 -20      0      0      0    I    0.0   0.0   0:00.00 kworker+
    5  root      0 -20      0      0      0    I    0.0   0.0   0:00.00 kworker+
    6  root      0 -20      0      0      0    I    0.0   0.0   0:00.00 kworker+
    7  root      0 -20      0      0      0    I    0.0   0.0   0:00.00 kworker+
    8  root      0 -20      0      0      0    I    0.0   0.0   0:00.00 kworker+
    9  root      20   0      0      0      0    I    0.0   0.0   0:00.00 kworker+
   10  root      20   0      0      0      0    I    0.0   0.0   0:00.01 kworker+

```

LOG #32 27/04/2025

Updated logbook.

LOG #33 30/04/2025

Team meeting – as per meeting minutes

LOG #34 1/05/2025

Menter/client meeting as per meeting minutes

LOG #35 2/05/2025

Catchup – made sure to review the documents in the OneDrive. Ensures that I am up to date.

LOG #36 4/05/2025

Updated logbook.

LOG #37 8/05/2025

Team Meeting – Ubuntu Test Execution

LOG #38 15/05/2025

Team Meeting – reviewed D-ITG data, assessed why we're getting inconsistent data.

LOG #39 17/05/2025

Observing and read the files that are in the OneDrive to ensure that I was up to date.

LOG #40 18/05/2025

Updated logbook.

LOG# 41 22/05/2025

Was unable to attend meeting – had to read the notes given from team members

- Mid progress report needs to be done ASAP
- Comparison of original proposal to the feedback given
- Team contribution
- New skills analysis based on upskilling

LOG #42 28/05/2025

Team meeting – I was assigned to do the meeting minutes.

- Discussion of status report.

LOG #43 28/05/2025

Filled out team contribution

LOG #44 29/05/2025

Meeting minutes completed – please refer to meeting minutes 28/05/2025.

LOG #45 29/05/2025

Catch up on the meeting missed – proofread the mid term proposal

LOG #46 1/06/2025

Updated logbook.

LOG #47 2/06/2025

Mid-term review = making sure to read the mid-term to have a full understanding.

LOG #48 8/06/2025

Updated logbook

LOG #49 13/06/2025

Mid-term prep = made sure to practice my section of the mid term prep. Also went through everything that is needed to prepare for questions.

LOG #50 15/06/2025

Updated logbook.

LOG #51 and 52 18/06/2025

Presentation + team-meeting after the review

Discussions – from the meeting minutes

Discussion on finding the theoretical reasons behind the results of evaluation

- How are the differences between TCP and UDP affecting the network performance?
- What is going on behind the scenes when running the evaluations?
- Documenting the differences between the IPv4 and IPv6
- How are the evaluation runs resulting in what we have in terms of the performance metrics?
 - o Average Delay, Jitter, Throughput and Packet loss
- How is the hardware affecting the results?
- What role does the operating system play in terms of the evaluation results and what is impacting the network performance?

SEMESTER 1 TOTAL = 113.27 hours.

SEMESTER 2

MEETING MINS: 28/08, 2/10

LOG #53

Meeting with Client = project progress update.

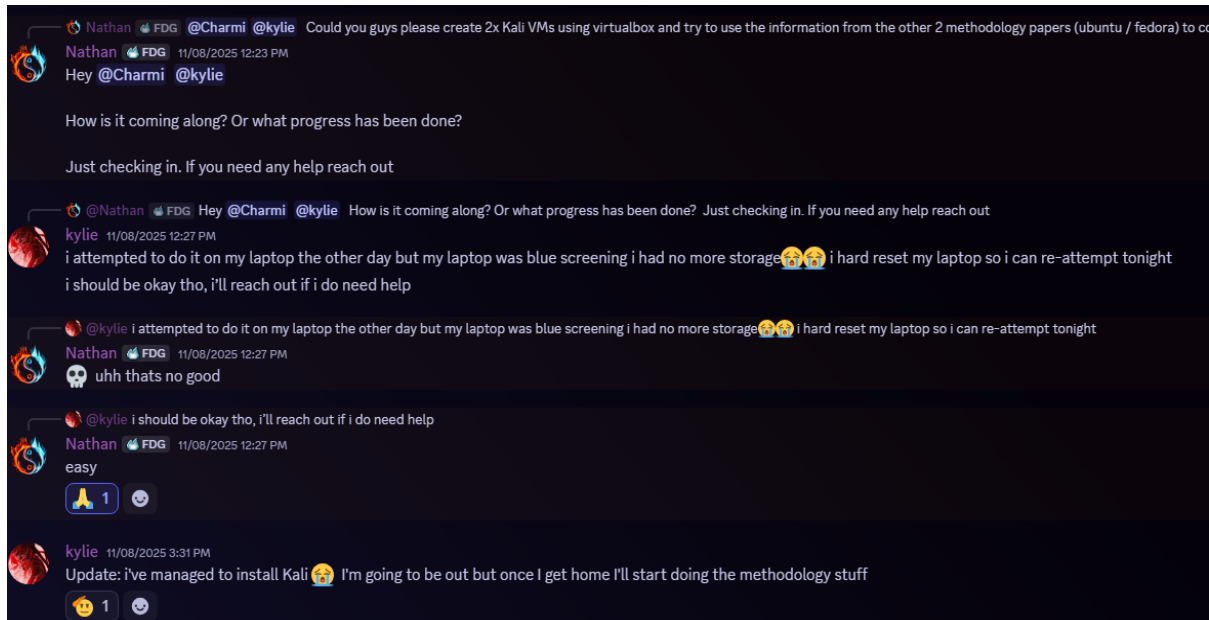
- Individual report
- Poster
- Client feedback
- Portfolio

LOG #54 & #55

Prep for Kali Installation and follow the methodology = install two vms onto VirtualBox.

However, my laptop had low a storage capacity and would blue screen – delaying the process.

I was unable to take a photo of the blue screen – but this is the chat where I explained the errors I was having.



https://www.youtube.com/watch?v=ZJFu0AoAY_g&t=615s = Video I followed, assisting me in the installation of the vms.

After installation = I followed the methodology given in the OneDrive to practice configuration skills.

I also collected important commands that I may need just in case:

- Ip addr show
- Ip link show
- Ip route show
- Route -n
- Ping
- Traceroute/tracepath.

LOG #56

Mentor/client meeting as per meeting minute.

LOG #57

Individual report = read through the document and planned out what I could possibly do.

Reviewed the assessment guideline

Also started section 1.

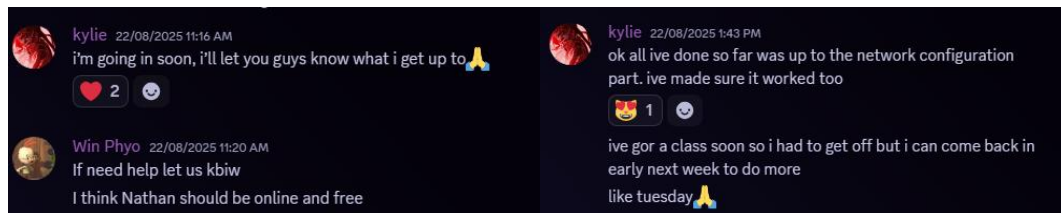
- Intro (maybe 200 words)
- Project overview and context
- Reflection on individual performance
 - o What happened
 - o What tasks did I do
 - o What went well.
 - o What improvements could I have made

LOG #58

Continued with the individual report from previous day.

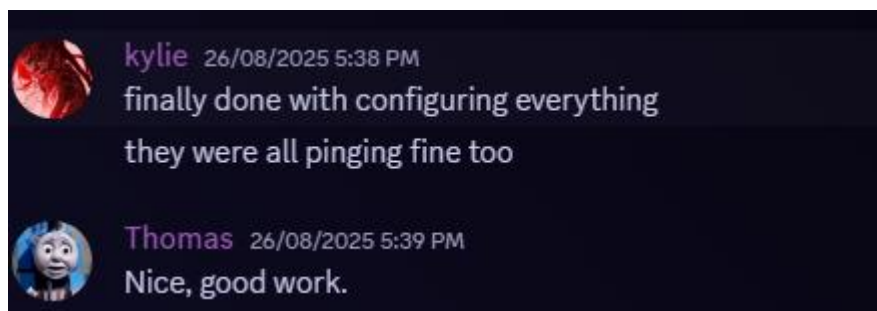
LOG #59

Went to configure Router 2 and followed the methodology. I was unfortunately unable to take photos but here are chat messages letting my team members know what I've done:



LOG #60

I did not take photos of when I was doing the Router 2 configuration onsite. However, I do have messages updating my team that me and Charmi have completed the configuration:



LOG #61

Attended workshop = attendance was monitored for the rest of the semester.

LOG #62

Meeting minutes filled:

As per meeting minutes document:

A2 Poster:

Take everything so far and compress it down to fit the poster.

Full of information with little text or words.

Task delegations will be discussed next team meeting:

- Getting everyone involved
- Refrain from using AI for any summarising

Team decided on finalising everything by at least week 10 to be fully prepared for mock presentation (Week 11 or 12)

Look into GitHub for version control and for our Portfolio folders.

Also, on setting up hyperlinks for documents.

LOG #63

Kali runs = Win taught me and Charmi how to do the runs and send it to a USB to convert the data into graphs.

LOG #64

Client meeting – discussion of Kali runs (refer to meeting minutes)

LOG #65

Attended workshop

LOG #66, 67, 69 and 70.

Linux from scratch = was assigned by Thomas to learn Linux from Scratch as it is essential.

Here are screenshots of me whilst doing LFS and also errors I was trying to resolve along the way due to incorrect commands:

```
vboxuser@UbuntuLFS: ~
passwd: password updated successfully
vboxuser@UbuntuLFS:~$ sudo shown -v lfs $LFS
sudo: shown: command not found
vboxuser@UbuntuLFS:~$ sudo chown -v lfs $LFS
chown: missing operand after 'lfs'
Try 'chown --help' for more information.
vboxuser@UbuntuLFS:~$ sudo chown -v lfs $LFS
chown: missing operand after 'lfs'
Try 'chown --help' for more information.
vboxuser@UbuntuLFS:~$ export LFS=/mnt/lfs
vboxuser@UbuntuLFS:~$ sudo chown -v lfs $LFS
changed ownership of '/mnt/lfs' from root to lfs
vboxuser@UbuntuLFS:~$ sudo chown -v lfs $LFS/*
chown: cannot access '/mnt/lfs/*': No such file or directory
failed to change ownership of '/mnt/lfs/*' to lfs
vboxuser@UbuntuLFS:~$ sudo blkid /dev/sdb1
/dev/sdb1: UUID="a373eecd-af5c-4865-887e-d51073a027c1" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="d7fe1f31-01"
vboxuser@UbuntuLFS:~$ sudo umount /mnt/lfs
umount: /mnt/lfs: not mounted.
vboxuser@UbuntuLFS:~$ sudo mount -o
mount: /mnt/lfs: can't find UUID=PUT-UUID-HERE.
vboxuser@UbuntuLFS:~$ df -h | grep lfs
vboxuser@UbuntuLFS:~$
```

```
/loop8: 576 KiB, 589824 bytes, 1152 sectors
ectors of 1 * 512 = 512 bytes
ize (logical/physical): 512 bytes / 512 bytes
(minimum/optimal): 512 bytes / 512 bytes
vboxuser@UbuntuLFS:~$ sudo mkfs.ext4 /dev/sdb1
mkfs2fs 1.47.0 (5-Feb-2023)
The file /dev/sdb1 does not exist and no size was specified.
vboxuser@UbuntuLFS:~$ sudo mkfs.ext4 /dev/sda2
mkfs2fs 1.47.0 (5-Feb-2023)
/dev/sda2 is mounted; will not make a filesystem here!
vboxuser@UbuntuLFS:~$
```

Device	Start	End	Sectors	Size	Type
/dev/sda1	2048	4095	2048	1M	BIOS boot
/dev/sda2	4096	52426751	52422656	25G	Linux filesystem

```
Device      Start    End      Sectors  Size    Type
/dev/sda1   2048     4095     2048     1M      BIOS boot
/dev/sda2   4096     52426751 52422656 25G     Linux filesystem

Disk /dev/loop8: 576 KiB, 589824 bytes, 1152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
vboxuser@UbuntuLFS:~$ sudo mkfs.ext4 /dev/sdb1
mkfs2fs 1.47.0 (5-Feb-2023)
The file /dev/sdb1 does not exist and no size was specified.
vboxuser@UbuntuLFS:~$ sudo mkfs.ext4 /dev/sda2
mkfs2fs 1.47.0 (5-Feb-2023)
/dev/sda2 is mounted; will not make a filesystem here!
vboxuser@UbuntuLFS:~$
```

```
vboxuser@UbuntuLFS: ~
Try 'grep --help' for more information.
vboxuser@UbuntuLFS:~$ mount | grep " /mnt/lfs "
vboxuser@UbuntuLFS:~$ sudo groupadd lfs
vboxuser@UbuntuLFS:~$ sudo useradd -s /bin/bash -g lfs -m -k /dev/
sudo: useradd: command not found
vboxuser@UbuntuLFS:~$ sudo useradd -s /bin/bash -g lfs -m -k /dev/
vboxuser@UbuntuLFS:~$ sudo passwd lfs
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
vboxuser@UbuntuLFS:~$ sudo shown -v lfs $LFS
sudo: shown: command not found
vboxuser@UbuntuLFS:~$ sudo chown -v lfs $LFS
chown: missing operand after 'lfs'
Try 'chown --help' for more information.
vboxuser@UbuntuLFS:~$ sudo chown -v lfs $LFS
chown: missing operand after 'lfs'
Try 'chown --help' for more information.
vboxuser@UbuntuLFS:~$
```

Linux from Scratch videos:

Told Thomas that LFS was extremely time consuming and suggested if watching tutorials on LFS was okay.

Got the confirmation I needed and watched LFS from this playlist:

<https://www.youtube.com/watch?v=A0tnO3OB4AY&list=PLyc5xVO2uDsDzdT8lkx430hZ-gY69wgS3>

Commands from previous upskilling and also whilst doing LFS

Linux commands:

pwd = print working directory; shows our current path

cd = change directory; changes directories

e.g. cd /home, cd (takes you to default directory).

man (command) = shows the manual for that command, sometimes some commands don't have manuals

ls = list folders

ls -a = shows all folders whether they are hidden or not.

ls -l = shows more details about the folders.

ls -la = basically combines -a and -l command.

ls -h = makes the size into a readable format.

You can hide files by adding a "." at the start of the name file.

Clear = clears the terminals

Mkdir = makes a directory

-p = parent directory. E.g. mkdir -p test/test/test (keeps creating folders).

Rmdir (directory) = removes the directory. You can't remove a directory if it isn't empty

Rmdir -p = removes the whole directory tree. You must specify the files in there.

Rm file.txt = removes a file (rm -r folder/ deletes a folder)

Touch (file.txt) = create an empty file.

Cp source target = copy files

COPY A FILE INTO THE SAME FOLDER (RENAME IT)

cp file1.txt file2.txt = makes a copy of file1 called file2 in the same directory

COPY AND RENAME AT THE SAME TIME

cp file1.txt /home/username/documents/file_backup.txt = copies file1 into documents and renames it.

COPY MULTIPLE FILES INTO A FOLDER:

file1.txt file2.txt file3.txt /home/username/desktop/ = all three files go into the desktop folder

COPY A FOLDER (NEEDS -R)

cp -r myfolder/ /home/username/backups/ = -r means recursive. Copy everything inside myfolder into backups.

COPY WITH OVERWRITE PROTECTION:

cp -i file1.txt /home/username = -i means interactive. If a file with the same name exists, linux will ask before overwriting.

COPY AND SHOW PROGRESS:

cp -v file1.txt /home/username/ = -v means verbose. It will print out what's being copied.

Pro tip = cp -riv myfolder/ /home/username/backups/ = you can combine the commands.

Mv old new = move/rename files

e.g. mv test2 test3. If you want to move back do mv ..

VIEWING AND EDITING FILES:

- Cat file.txt = display contents of a file.
- Less file.txt = view file one page at a time.
- Nano file.txt or vim file.txt = edit a file in terminal.
- Head file.txt = first 10 lines
- Tail file.txt = last 10 lines (tail -f logfile.log follows a live updating log).

System Info and Monitoring:

- Uname -a = system info (kernel, architecture)
- Top = live view of processes and resource usage.
- Htop = better version of top (if installed)
- Free -h = memory usage
- Df -h = disk usage
- Uptime = how long the system has been running.

Permissions and Ownership:

- Ls -l = shows file permission (rwx).
- Chmod 755 file.sh = change permission (7=read/write/execute)
- Chown user:group file.txt = change ownership
- Whoami = shows your current user.

Networking:

- Ifconfig or ip addr = shows network interfaces
- Ping google.com = test connectivity
- Curl example.com or wget example.com = fetch content from the web
- Netstat -tulnp or ss -tulnp = check open ports and services

User Management:

Adduser username = create a new user

Passwd username = set/change password

Su - username = switch user

Sudo = runu command as admin/root

LOG #68

Showed Raymond our Kali graphs – also looked at the draft of our poster.

- Me and Charmi in charge of explaining the data we collected. Talked about how the data seems to follow the same trends
- However, has fluctuations in certain runs.
- Ubuntu was the best statistically

LOG #71 and 72:

Individual report = started section 2.

Drafted possible layouts for how I can write this up.

Purpose = explain what this report is about

How it helps with my growth throughout the project, learning, contributions.

Challenges.

Scope of reflection:

Areas that I need focus on e.g. planning, communicating etc.

Learning outcomes:

What have I learnt throughout this project?

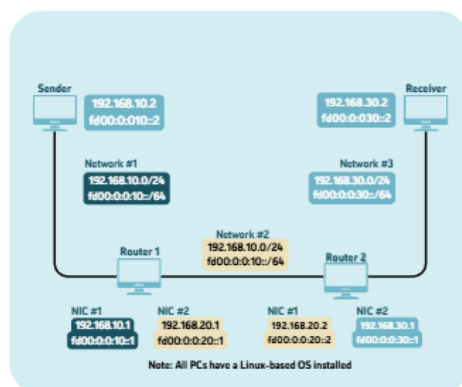
How did this help me improve?

Conclusion.

LOG# 73

Poster = me and Larissa were in charge of creating the poster, summarizing and fitting everything into the poster.

Had to recreate the topology to follow the theme.



layriceha started a call that lasted 7 hours. 14/10/2025 11:09 AM

Summarized Background, rationale, objectives, project impact and goals to fit into the poster:

Background:

The Internet Protocol continues to evolve to meet growing global connectivity demands, but IPv4's limits are making it an hindrance (Ugwumba & Jaja, 2025).

IPv6, its intended replacement, offers more addresses (≈ 4.3 billion vs. ≈ 340 undecillion), stronger security features, and improved routing efficiency (Amazon, 2025).

Our client requested an evaluation of the network performance of three Linux-based operating systems configured as routers, to determine which performs best under both IPv4 and IPv6, and to compare the results with his previous evaluations.

Rationale:

Limitations in real-world data and IPv4's scalability constraints make it difficult to transition to IPv6 (Narayan et al., 2016). A comparison of IPv4 and IPv6 performance will help us understand the implications of migrating to IPv6.

Objectives:

Network performance tests compared Ubuntu, Fedora, and Kali Linux as software routers in a 4-computer daisy-chain network topology (2 clients, 2 routers) (Figure 2).

The evaluation will:

- Generate TCP/UDP traffic over IPv4 and IPv6 using D-ITG.
- Measure throughput, delay, jitter, and packet loss.
- Test with packet sizes ranging from 128 bytes to 1536 bytes.
- Requires a minimum of ten runs per evaluation.

Project Impact:

This evaluation provides real-world data that compares the Network Performance between the different Linux operating systems and IPv4 and IPv6.

- Improves Understanding: Understands the pros and cons of transitioning to IPv4 & IPv6 and differences with TCP/UDP.

The results allow Dr. Lutui to compare historic and future data to track performance trends influencing IPv6 adoption on Linux.

- Future Protocol Application: Aims to encourage IPv6 adoption and support its transition into newer protocols like QUIC.

Goals:

Compare IPv4/IPv6 performance: Achieved using TCP/UDP to find key differences and support the future transition to IPv6.

Provide Results: Supports Dr. Lutui in analysing trends for future networking development.

Upskilling: Strengthened Linux and Networking knowledge.

Communication: Consistently keep members updated.

Improvements: Consistent feedback from Client and Supervisor for continuous improvement (as per STLC methodology).

LOG #74

Team meeting as per meeting minutes = ubuntu test execution.

NOTES FROM MEETING MINS:

#	Discussion and Decisions
1.	IPv4 Team – running Ubuntu evaluations and logging them. Transferring the logs to our personal computers for IPv6 Team to graph.
2.	Proposal Feedback Recapping to start working on the Midterm review status report Recalling Proposal Feedback from Mentor and Moderator: • Follow STLC methodology phases as it is, • Ensure the Schedule shows the iteration or some sort of way to tell that we would go back and forth through the phases, • Ensure that the Gantt Chart incorporates the technical aspect in all phases in Entry/ Exit Criteria, • The schedule should have the execution phase started before the end of first trimester, • Suggestion: While IPv4 does the execution, IPv6 can be upskilling. Vice versa., • Some of the out of scope are within the scope (Assuming some of bullet points listed in the slide)
3.	Delegated Team members to work on Fixing the Proposal Feedback

LOG #75

As per meeting minutes.

Mock presentation = most of all the sections that were in our proposal just needed to be kept short. A lot of the paragraphs had to be simplified into 2-3 sentences.

LOG #76

Team meeting – debriefing what happened in the mock presentation

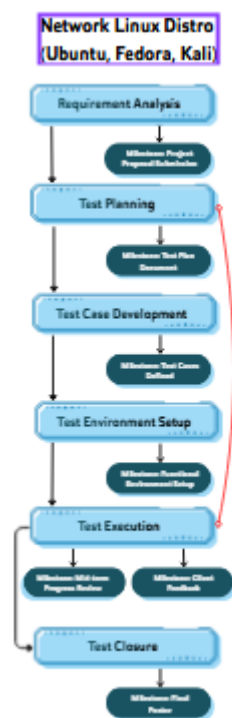
Delegated tasks for changing the format of the Final Poster sections and layout.

Planned date for printing Final Poster: Monday NOV 3

LOG #77

Inaccurate runs so we had to collect data again by going on-site and running the PCs. This was done with Win and Charmi, Win helped us understand what each command did.

LOG #78 and 79



Had to recreate on Canva to follow Daniel's feedback.

Suggested would be better if stair-cased down as followed.

Also proofread the final poster and gave suggestions with the design.

LOG #80

Worked with the QA Phase entry/documentation with Win, Nathan and Larissa.

LOG #81

Attended workshop

LOG #82:

Individual report has not been completed. Applied for SCA and is due Friday.

Started to proofread what I have got to ensure that everything made sense and was accurate.

LOG #83

REVIEWED LOGBOOK ONE MORE TIME.

TOTAL HOURS FOR BOTH SEM = 206.02